

assignment _equilibrium

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:57pm

Quiz Instructions

There is tutorial for most of the questions. HIDE the hints for the other questions

Try first on your own. These are not easy questions. take your time. You can get an extension if you need,



Question 1 6.7 pts

An object is being acted upon by three forces and as a result moves with a constant velocity. One force is 60.0 N along the +x-axis, and the second is 75.0 N along the +y-axis. What is the magnitude of the third force?

☐

15.0 N

☐

96.0 N

☐

67.5 N

☐

135 N

☐

48.0 N



Question 2 6.7 pts

A 400-kg box is lifted vertically upward with constant velocity by means of two cables pulling at 40.0° on either side of the vertical direction. What is the tension in each cable?

☐

3920 N

☐

2560 N

☐

231 N

☐

400 N

☐

800 N



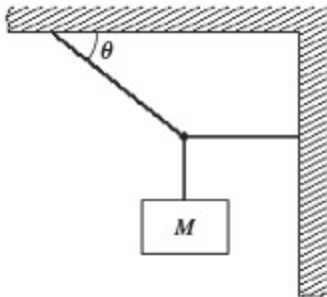
Question 3 6.7 pts

If I weigh 741 N on Earth at a place where $g = 9.80 \text{ m/s}^2$ and 5320 N on the surface of another planet, what is the acceleration due to gravity on that planet?

81.0 m/s^2 61.2 m/s^2 51.4 m/s^2 70.4 m/s^2 

Question 4 6.7 pts

The figure shows a block of mass M hanging at rest. The light wire fastened to the wall is horizontal and has a tension of 38 N. The wire fastened to the ceiling is also very light, has a tension of 59 N, and makes an angle θ with the ceiling. Find the angle θ .



50°



40°



65°



33°



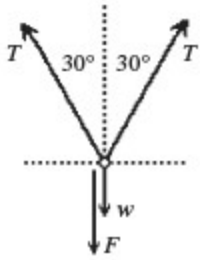
45°



Question 5 6.7 pts

A 6.00-kg ornament is held at rest by two light wires that form 30° angles with the vertical, as shown in the figure. An external force of magnitude F acts vertically downward on the ornament. The tension exerted by each of the two wires is denoted by T . A free-body diagram, showing the four forces that act

on the box, is shown in the figure. If the magnitude of force F is 410 N, what is the magnitude of the tension T ?


☐

376 N

☐

188 N

☐

235 N

☐

271 N

☐

470 N

⋮

Question 6 6.7 pts

What is the mass of an object that experiences a gravitational force of 685 N near Earth's surface where $g = 9.80 \text{ m/s}^2$?

☐

69.9 kg

☐

72.7 kg

☐

71.3 kg

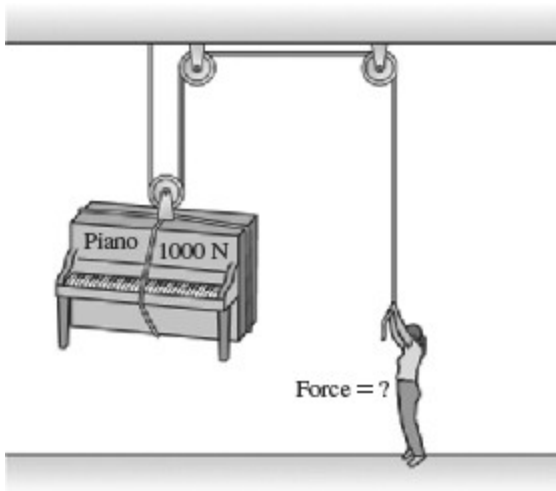
☐

68.5 kg

⋮

Question 7 6.7 pts

A piano mover raises a 1000-N piano at a constant speed using a very light rope in a frictionless pulley system, as shown in the figure. With what force is the mover pulling down on the rope?



☐ Depends on the speed of the piano.

☐ 250 N

☐ 1000 N

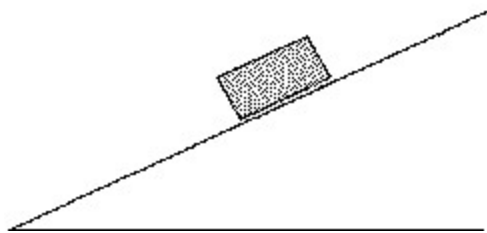
☐ 500 N

☐ 2000 N



Question 8 6.7 pts

A brick is resting on a rough incline as shown in the figure. The friction force acting on the brick, along the incline, is



☐ greater than the weight of the brick.

☐ equal to the weight of the brick.

☐ zero.

☐ less than the weight of the brick.

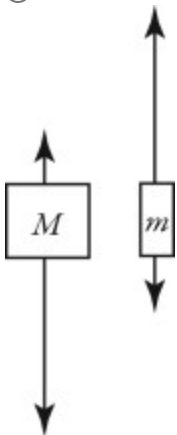


Question 9 6.7 pts

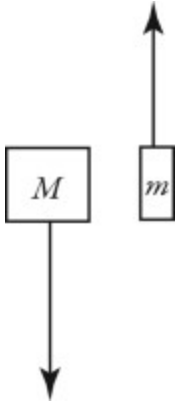
Two unequal masses M and m ($M > m$) are connected by a light cord passing over a pulley of negligible mass, as shown in the figure. When released, the system accelerates. Friction is negligible. Which figure below gives the correct free-body force diagrams for the two masses in the moving system?



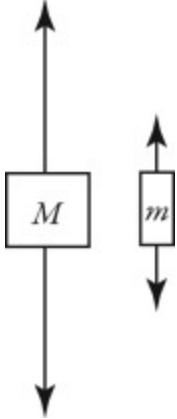
hint: since $M > m$ then M is accelerating down and m is accelerating up. They both have the same acceleration in magnitude. If there is an acceleration, there is a net force. First consider the mass M . (only this system). it is accelerating down. so the down force (weight) has to be larger than the up force (tension). consider the small mass m . it is accelerating up. So the up force (tension) has to be larger than the down force (weight)

☐


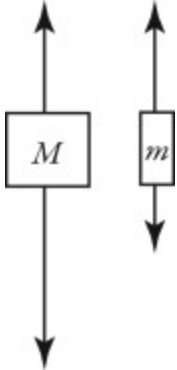
○



○



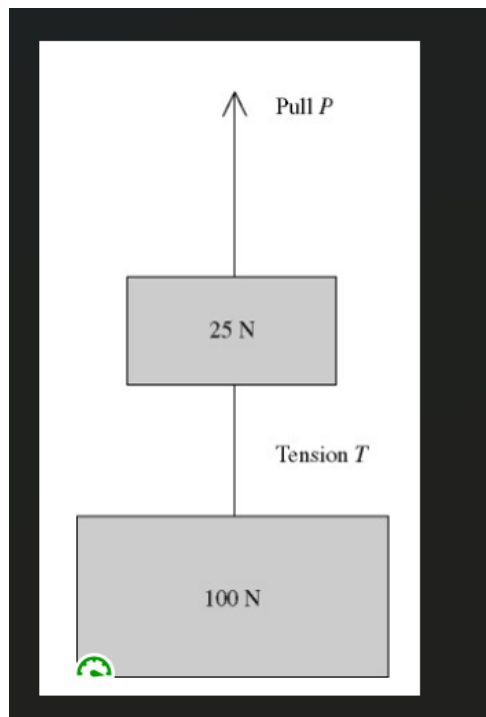
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⋮

Question 10 6.7 pts

Two weights are connected by a massless wire and pulled upward with a constant speed of 1.50 m/s by a vertical pull P . The tension in the wire is T (see figure). Which one of the following relationships between T and P must be true?



hint: The pull P has to carry the 2 blocks. Place the 2 blocks in an imaginary bag. $P = \text{total weight} = 125\text{N}$

The tension T has to carry the 100N block only. $T = 100$

☐

$$P = T + 25 \text{ N}$$

☐

$$P = T + 100 \text{ N}$$

☐

$$P + T = 125 \text{ N}$$

☐

$$T > P$$

☐

$$T = P$$

☒

Question 11 6.7 pts

If I weigh 824 N on Earth at a place where $g = 9.80 \text{ m/s}^2$ and 5320 N on the surface of another planet, what is the acceleration due to gravity on that planet?

☐

$$72.8 \text{ m/s}^2$$

☐

$$55.1 \text{ m/s}^2$$

☐

$$46.2 \text{ m/s}^2$$

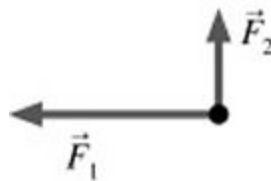
63.3 m/s²

Question 12 6.7 pts

Figure 5.1 shows two forces acting on an object. They have magnitudes $F_1 = 9.6$ N and $F_2 = 3.2$ N. What third force will cause the object to be in equilibrium?

hint: we did a very similar one in class. You can see that F_3 is in the 4th quadrant. Its like the lab you did. Geometrically, add F_1 and F_2 (tail to head) then F_3 is in the opposite direction with the same magnitude. See my slide.

Figure 5.1

6.4 N at 162° counterclockwise from $\vec{F}_1$ 6.4 N at 108° counterclockwise from $\vec{F}_1$ 10 N at 162° counterclockwise from $\vec{F}_1$ 10 N at 108° counterclockwise from $\vec{F}_1$ 

Question 13 6.7 pts

Figure 5.2 shows two forces of equal magnitude acting on an object. If the common magnitude of the forces is 2.3 N and the angle between them is 40°, what third force will cause the object to be in equilibrium?

hint: Have the x-axis bisecting the sector F_1, F_2 . (x-axis is the axis of symmetry). Then F_1 makes an angle of 20 with (-x-axis). and F_2 also makes an angle of 20 with (-x-axis)

$F_{1y} + F_{2y} = 0$ shows that $F_1 = F_2 = F$ in magnitude

$$F_{1x} + F_{2x} + F_{3x} = 0$$

Figure 5.2

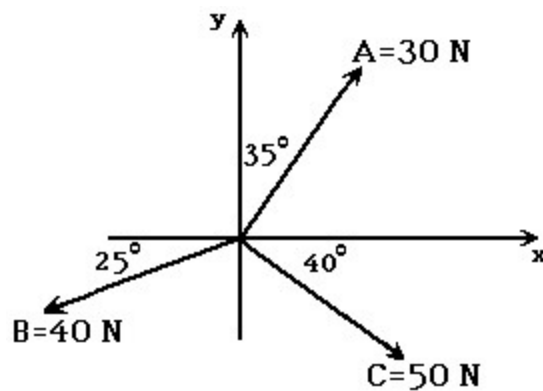


- ☐ 1.8 N pointing to the right
- ☐ 3.5 N pointing to the right
- ☐ 2.2 N pointing to the right
- ☐ 4.3 N pointing to the right



Question 14 6.7 pts

Figure 5.5



Three forces A, B, and C act on a body as shown. A fourth force F is required to keep the body in equilibrium.

In Figure 5.5, the x-component of force F is closest to:

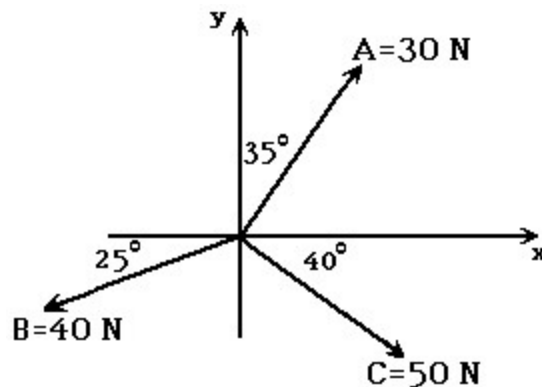
hint: $A_x + B_x + C_x + F_x = 0$ then solve for F_x . $A_x = 30\cos(55)$, $B_x = 40\cos(105)$ etc....

- ☐ -32 N
- ☐ +28 N
- ☐ -19 N
- ☐ -28 N
- ☐ +19 N



Question 15 6.7 pts

Figure 5.5



Three forces A, B, and C act on a body as shown. A fourth force F is required to keep the body in equilibrium.

In Figure 5.5, the y-component of force F is closest to:

hint: $A_y + B_y + C_y + F_y = 0$ solve for F_y

☐

-32 N

☐

+28 N

☐

+24 N

☐

+32 N

☐

-28 N

Not saved

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